

Major Project Report
on
Moist air condensation enhancement via superhydrophobic surfaces
Submitted to
Amity University Uttar Pradesh



In partial fulfillment of the requirements for the award of the degree
of
Bachelor of Technology
in
Mechanical Engineering
by
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Under the guidance of
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DEPARTMENT OF MECHANICAL ENGINEERING
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Declaration

In partial fulfillment of the requirements for the award of the Bachelor of Technology in Mechanical Engineering degree, I, **Apoorva Singh**, a student of **B.Tech (8 ME-1(Y))**, hereby affirm that the project I submitted to the Department of Mechanical Engineering, Amity School of Engineering Technology, Uttar Pradesh, Noida, under the title "**Moist air condensation enhancement via superhydrophobic surfaces.**" has never before served as the foundation for the award of any degree, diploma, or other title or recognition of a similar nature. With the exception of a few brief passages that only call for appropriate recognition in academic writing, the author attests that permission has been acquired for the use of all copyrighted material contained in the dissertation or project report, and that all such usage is acknowledged.

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(8 ME-1(Y)) (2020-2024)

CERTIFICATE

On the basis of a declaration submitted by **Mr. Apoorva Singh**, student(s) of B. Tech Mechanical Engineering, I hereby certify that the project titled “**Moist air condensation enhancement via superhydrophobic surfaces**” which is submitted to Department of Mechanical Engineering, Amity University Uttar Pradesh, in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology Mechanical Engineering, is an original contribution with existing knowledge and faithful record of work carried out by him/them under my guidance and supervision. To the best of my knowledge this work has not been submitted in part or full for any Degree or Diploma to this University or elsewhere.

Noida

Date

Signature of Guide

Dr. Basant Singh Sikarwar

Guide

Department of Mechanical Engineering

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Acknowledgement

I extend my heartfelt thanks to Prof. (Dr.) **Basant Singh Sikarwar**, who serves as both the Head of the Department of Mechanical Engineering and my dedicated faculty guide at Amity University, for entrusting me with this project. Their steadfast assistance and direction were crucial to this project's successful conclusion.

I also wish to express my appreciation to my fellow batch mates for their assistance, ideas, and motivation.

Apoorva Singh.

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Nomenclature

H_{lv}	Condensation Enthalpy, $Wm^{-2}K^{-1}$
Q	Heat Transfer Rate, W
m_a	Mass flow rate of air, kgs^{-1}
m_r	Mass flow rate of refrigerant, kgs^{-1}
ω	Specific humidity, kg/kg
P_1	Inlet Pressure of compressor, bar
P_2	Outlet Pressure of compressor, bar
h	Heat transfer coefficient, $Wm^{-2}K^{-1}$
h_w	Heat transfer coefficient of water, $Wm^{-2}K^{-1}$
T	Temperature, $^{\circ}C$
C_p	Specific Heat in constant pressure, $Jkg^{-1}K^{-1}$
I	Exergy destruction, J/kg

Abbreviations

VCRS	Vapor Compression Refrigeration Cycle
AWG	Atmospheric Water Generator
VC-AWG	Vapor Compression Atmospheric Water Generator
MHI	Moisture Harvesting Index
PFOCTS	Perfluoro-Octyl-Trichlorosilane
CDAQ	Compact Data Acquisition
Rh	Relative Humidity

Subscripts

0	Ambient conditions
comp	Compressor
cond	Condenser
evp	Evaporator
exp	Expansion device
1, 2, 3, 4, 5, 6, 7, 8	Inlets and Outlets of different components

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Introduction

Approximately 25% of the global population, equivalent to around 3.5 billion people, faces drinking water crisis, despite approximately 75% of the earth's surface being covered by water[1,2]. Ground water, lakes, and rivers are conventional sources of drinking water. These existing water resources are being depleted in many areas due to climate changes and exponential population growth and inadequate treatment of saline water[3,4]. In addition, drinking water consumption has increased globally by ~1% per year over the last 40 years and is expected to grow at a similar rate through 2050[5]. Therefore, other sources and methods to get fresh drinking water need to be explored. Water desalination is one of the methods of alternative sources of water in countries and regions that have access to sea water[6,7]. Hence, it is economically viable for coastal regions. In contrast, atmospheric water harvesting is a potential source of water because the atmosphere contains water vapor around $1.29 \times 10^{13} \text{ m}^3$ and this water vapor is ubiquitous and adequate to resolve water crises forever[8,9]. In addition, it is acquired without any restrictions of geographical and hydrologic conditions, and it can be used safely without any purification. In this technique moist air is dehumidified when surface temperature (T_s) is below the dew point temperature (T_d). But large cooling power is required where atmospheric conditions are adverse such as low temperature and low humidity. Hence, degree of subcooling ($T_d - T_s$) is essential for moist air dehumidification process and cooling feasibility of atmospheric water generator (AWG) is evaluated by Moisture Harvesting Index (MHI)[4],

$$\text{MHI} = \frac{H_{lv}}{Q} \quad (1)$$

Here, H_{lv} condensation enthalpy and Q is the total given heat. Hence, hot and humid regions as the tropical areas of the earth require less energy for the dehumidification process. On the other hand, many studies improve atmospheric water harvesting capability by altering morphology and wettability of condensing surfaces[10–13]. In addition, many scientists used sorbent-assisted vapor capture as a type of pretreatment of condensation that concentrates the vapor and makes the subsequent cooling and condensation more efficient[7,8,14–16]. An excellent AWG system have combination of these three techniques[7,17–20] (a) combination passive and active cooling techniques

for efficient cooling, (b) altering surface morphologies for efficient condensation rate and (c) realizing sorbent-assisted system for improving the moisture content for condensation. In passive cooling of AWG, the degree of subcooling is completely driven by natural and sustainable power[21,22]. One of the examples of passive AWG is fog and dew collection on large-area nets and surfaces by radiation cooling. In this technique, water collection is limited by environmental conditions. In the active cooling for AWG technique[23–25], the desired cooling of the condensing surface is achieved by electricity such as vapor compression cooling and thermoelectric coolers. It has an excellent output of water yield ranging from 10-50 liters/hour. But active cooling is costly and has adverse environmental effects in terms of carbon footprint. Many studies [9,16] used active, passive and combination of them for improving the yield of harvested water. Singh et al. [26] reported that the rate of condensation increases as the degree of subcooling increases.

In this work, AWG was designed and fabricated which has an effective active cooling method and altered morphology copper tubes as condensing surfaces. Therefore, reviewed literature based on the cooling method and altering the condensing surface morphologies.

In literature[17], vapor compression and peltier devices are used as active cooling and are widely used for water harvesting from the atmosphere. Farayedhi et al.[23] used a vapor compression system for condensing water vapor of moist air in hot and humid regions. In their experimental study, they installed 1.5-ton split air conditioning in Dhahran, Saudi Arabia for cooling and dehumidification of air at different ranges of dry bulb temperatures and relative humidity. They reported that large production of water occurs at large relative humidity and maximum capacity of water prediction 2.50 kg/ton per CDD. Farshid Bagheri [22] investigated the performance of commercial atmospheric water harvesting systems and his studies showed that water harvesting rate, energy intensity ranging from 1.02 kWh/L for warm and humid climates to 6.23 kWh/L for cold and humid climates. Badr A. Habeebullah[18] study concluded that using evaporator coils as water collection surfaces in hot and humid areas strongly affects the air velocity and water yield decreases with increasing air speed due to insufficient evaporator thermal capacity.

Despite all the shortcomings, thermoelectric coolers are widely used for water

harvesting from the atmosphere[13]. Shourideh et al.[14] used a thermoelectric cooler for condensing water vapor of moist air and perform a comparative study with other AWGs using similar systems, tests were conducted in a contained environment with ambient temperature maintained between 30-35°C and 60-80% relative humidity. They reported a water generation rate of 32-66 ml/hr with a low power consumption of 60 W. Vián et al.[15] designed a two-stage thermoelectric powered dehumidifier with optimized voltage supply and his studies concluded the coefficient of performance of the device lies between 0.7-1.3 while offering real-life applications of such devices despite relatively lower COP as compared to its vapor-compression counterpart. Tan and Fok et al.[16] developed a thermoelectric cooler with a cold box that assists in condensation; they were able to record an average power consumption of 89 W, with condensation rates at 5340 kWh/kL. Despite such poor performances, a proposal for improved prototypes has been kept forth. Joshi et al.[17] studies concluded that the water condensate is directly proportional to the humidity of the moist air as the water vapour content in the same mass of air increases. The water condensate is also directly proportional to the electric current given to the peltier modules because of the increase in heat absorbed by the modules. As such, the practicality of devices can be increased by supplying electricity using solar panels, as was accomplished by Moria et al.[18].

In literature, Copper and Aluminum metallic surfaces were basically used as condensing surfaces[27,41]. The nature of these surfaces is hydrophilic and water vapor condenses on these surfaces in the form of film which is known as filmwise condensation[28]. The condensation rate of filmwise condensation is at least one order smaller than the dropwise condensation. Hence, dropwise condensation is an efficient mode of condensation, and this efficient mode of condensation occurs when the condensing surface is hydrophobic[29]. The water vapor condenses on these surfaces in the form of droplets and more details of the dropwise condensation was explicitly reported elsewhere [30–33].

Coating low energy materials [34,39], and altering the surface morphology by chemical etching [35,42,43] and physical texturing [36,40,44] were very extensively used for fabricating hydrophobic surfaces of Cu and Al. These methods have their own pros and cons such as physical texturing is expensive, and it is also limited by realizing the size of the surface. However, low surface energy coating is easy and can be used any size of the

surface. But it has durability and sustainability problems in hot and humid environments. Therefore, hybrid techniques such as plasma etching with low surface energy coating are employed in this work for fabricating the condensing surface.

With this background, vapor compression cooling based atmospheric water generator (AWG) is designed and fabricated in this work. In this system, air flows over the cooling surface and it is cooled with dehumidification. The dehumidified cooled air flows over the compressor to minimize the compressor work via its cooling. Later, it flows over the condensing surface towards the atmosphere. In this way, the cooling effect of air is used for minimizing the power consumption of compressors and condensers. Experiments were carried out in Months of July to September in Noida, India. Because these days climate conditions (humidity is more than 50% and temperature is moderate of 28-35°C) are more favorable for condensation. These climate conditions occur in most of the coastal part of India. Our aim is to propose AWG one pragmatic device for the coastal part of India. Experiments were carried out from morning 10:00 AM to afternoon 4:00 PM. The temperature, humidity, air velocity, power consumption was recorded at intervals of 30 minutes from the commencement of experiment. Temperatures (eight locations of the system and pressure ratio (pressure at the compressor outlet and outlet of water vapor condensing surface) of the refrigerant cycle were also recorded for energy and exergy analysis. Based on the experimental data, the rate of water production and power consumption were estimated at various environmental conditions. In addition, climatic conditions such as temperature, pressure, and relative humidity for feasibility and affectability of the AWG are analyzed. This study can be helpful for designing parameters in terms of water harvesting threshold, energy consumption and water production cost.

A numerical analysis was also conducted using the solar based peltier-cooled AWGs. A thermoelectric cooling based atmospheric water generator (AWG) is designed and fabricated for analysis. This system consists of a series of ducts through which moist air passes over a treated aluminum plate cooled by a peltier device forming a condensate over the surface, the dehumidified air is then ejected through an exhaust fan. The droplets are collected in a measuring tank for testing, further, temperature and humidity was measured at different points of the apparatus. The study can be helpful for design and testing of such AWG through Computational Fluid Dynamics (CFD) analysis.

Experimental setup

a. Thermodynamic Analysis of Vapor Compressor Cooling Based Atmospheric Water Generator

Figure 1 (a-b) shows schematic diagram and photograph of experimental setup, respectively. The setup consists of a compressor, expansion device and multiple tubes and multiple passes tubes with fin arrangement two identical heat exchangers. These heat exchangers are known as heat exchanger#1 and heat exchanger#2. In the heat exchanger#1, the outer surface of tubes and their fins work as a condensing surface of moist air. In this heat exchanger, refrigerant flows inside the tube and moist air flows over tubes, moist air transfers its heat to the refrigerant via cooling with dehumidification. The heat exchanger#2 works to transfer heat from the refrigerant to the environment by flowing the cold and dehumidifying air over the tubes and fin surfaces. However, refrigerant flow inside the tube is condensed and transfers the heat to air. The unique feature of this device is to use the cold and dehumidified air to flow over the heat exchanger#2. This enhances the performance of the setup.

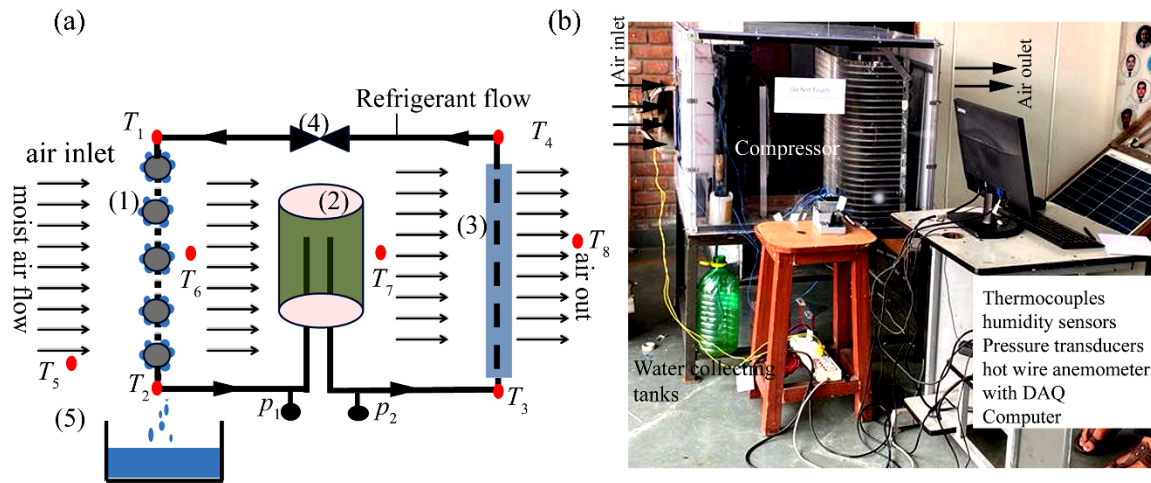


Figure 1: (a) Schematic diagram of Vapor compression based Atmospheric water generator (VC-AWG) with position of thermocouples for measuring temperature and (b) Photograph of experimental setup. ● Position of thermocouples for measuring the temperature at various locations. (1) Heat exchanger#1, (2), compressor, (3) Heat exchanger#2, (4) expansion device, (5) water collecting tank. (P_1 and P_2) pressure tapping points for measuring the pressure before and after flow of the compressor.

Heat exchanger#1 and Heat exchanger #2 are identical in shape and size and These heat exchangers were fabricated by one of the leading manufacturing industries. (ambergrouppindia.com) of air conditioning units in India. The dimensions and drawing of these heat exchangers are shown in Figure 2.

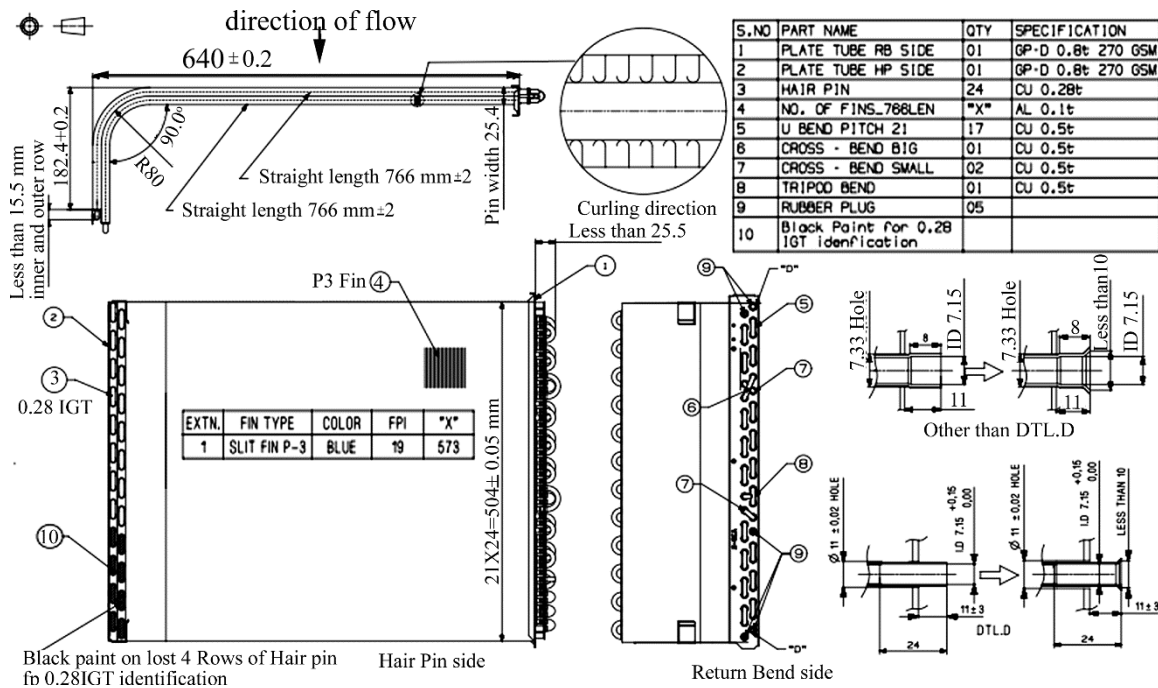
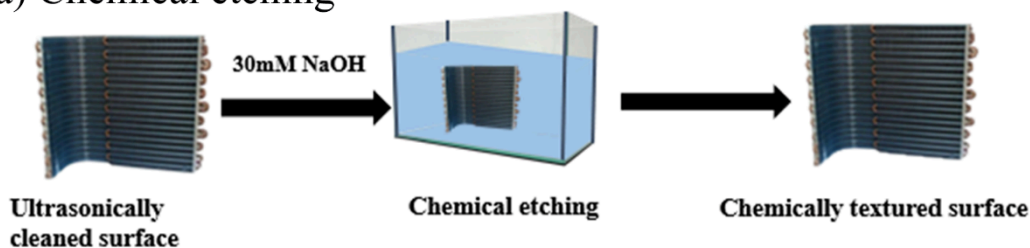


Figure 2: Drawing with dimensions of heat exchanger which is used in this device as evaporator and condenser.

The detailed specifications of all components with their photographs are given in table 1. The outer tubes and fin of heat exchanger#1 were made hydrophobic by chemical etching with a coating of low surface energy. Figure 3 shows a schematic diagram for fabricating the hydrophobic coating of PFOCTS. The heat exchanger is chemical etched before dipping in solution of PFOCTS. The measured contact angle of the water droplet on the surface of the tube is more than 150° . Two fans were fastened to circulate the moist air in controlled conditions over the tubes and direction of the flow is such a way that it first flows Heat exchanger#1 where it is going to process dehumidification and cooling. After cooling and dehumidification process, it flows over compressor and heat exchanger #2 and cools down the compressor and tubes of heat exchanger#2.

(a) Chemical etching



(b) Chemical coating

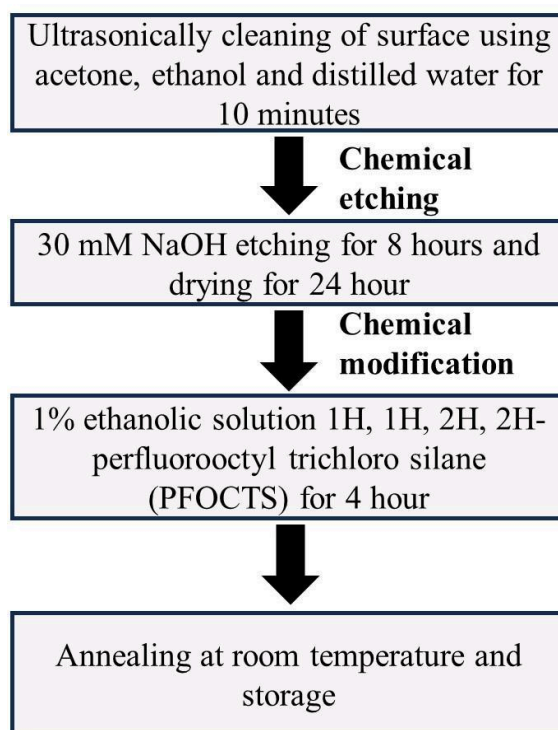
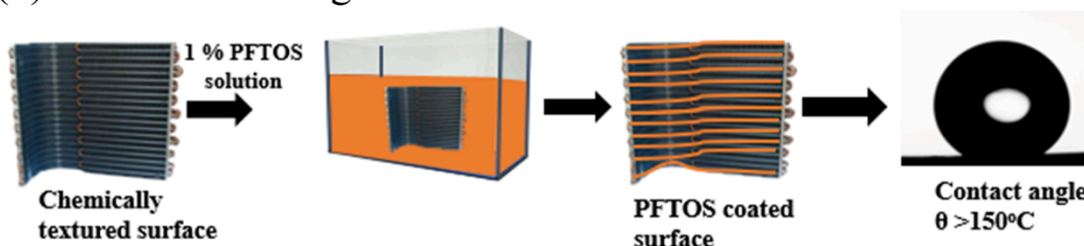
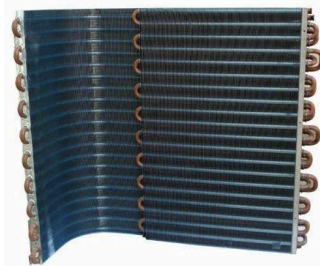


Figure 3: (a) Graphical representation of methods and material for fabrication of super-hydrophobic metallic surface and (b) Flow diagram of various steps of surface preparation.

Table 1: The components of experimental setup with their specification and photographs

Name of components with specification	Photographs
Compressor- -Model: Emerson CR30K6M -Item Weight: 30 Kg, Refrigerant: R-22 -Voltage: 230 V -Type: Single phase, 50 Hertz -Max. Power: 1750 W -Capacity: 1.5 tons -Cooling capacity: 25,000 Btu/h	
Heat exchanger#1 and heat exchanger#2 Specifications: There are multiple tubes and multiple passes, dimensions given in figure 2.	
Pressure gauge: -Nominal size: 63mm -Accuracy class: 1.0-1.6 -Scale ranges: 0-1 to 0-400 bar -Process connection (copper alloy): VCR®-compatible face seal fitting. -Permissible temperature range: -40-+100 °C	
Data Acquisition system with T-type thermocouple -Model: NI cDAQ-9174 -Number of counters/timers: 4 -Input FIFO size: 127 samples per slot -Timing accuracy: 50 ppm of sample rate -Timing resolution: 12.5 ns -Onboard regeneration: 16 -Waveform acquisition (DI) FIFO: 127 samples per slot -Waveform generation (DO) FIFO: 2,047 samples	
Water collection measuring beaker -Capacity: 500 ML -Material: Glass	

Velocity measurement system -Model: Metrix AVM-05 Anemometer -Circuit: Custom LSI microprocessor circuit -Temperature sensor: NTC-type thermo resistance -Operating Temperature: 0-50°C, < 80%RH -Weight: 0.57 Kg -Power Supply: 4x1.5V AAA batteries -Data storage: 24 groups -Air velocity range: 0.5-45 m/s -Air flow CMM: 0-999 m³/min -Sensor: conventional angled vane arms with low friction ball bearing	
Fan for air circulation- -Model: Havells Turbo Force -No: THE SPDB 129 -Speed: 900 RPM -No. of Blades: 4 -Type of Product: Exhaust Fan -Sweep Size: 300 mm -Voltage: 230V -Phase: Single Phase -Power: 50 W	
Humidity measurement device -Model: KUSAM-MECO KM-918 Thermo Hygrometer -Operating temperature: 0-55°C, <75% R.H. -Power Supply: 9V Battery -SKU: ME.TE.DI.160285 -Measurement Rate: 0.5 times/s -Weight: approx. 320g -Dimension: 178(H)x65.5(W)x35(D)mm	

Condensed water over the fins and tubes of heat exchanger#1 is collected in the tank during the experiment. T-type thermocouples are used to measure temperature in eight locations and this temperature was recorded every twenty minutes on the computer using CDAQ-9174. Air flow through the fans was controlled by the flow regulator of fans and their flow was measured by turbine type air flow meter. The refrigerant pressure at inlet (P_1) and outlet (P_2) of the compressor were measured by pressure gauge arrangement.

Using the surface temperature of tubes at various locations and pressure difference created by the compressor, energy analysis is carried out for each component of the vapor compressor cycle. Figure 4 shows the mass and energy balance of the control volume of each component. Figure 4(a) shows the cooling of dehumidification of moist air flow over cooling and fins of Heat exchanger#1. However, mathematical expressions

of mass and energy balance were given in Equations (1-5).

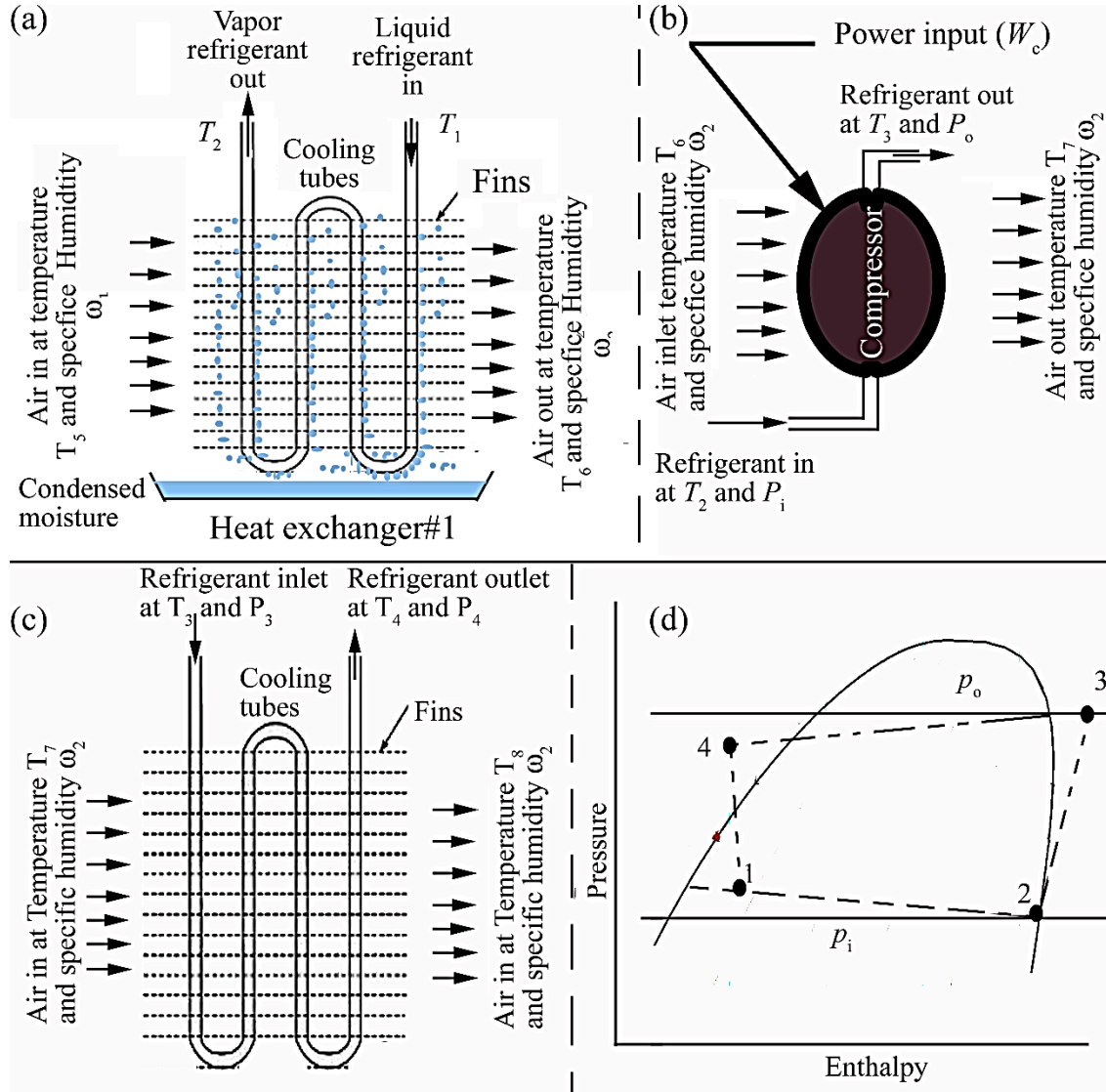


Figure 4: Schematic diagram of various component of AWG and Pressure enthalpy diagram for mass and Energy balance (a) cooling tubes with fin of Heat exchanger#1, (b) compressor, (c) Heat exchanger#2 and (d) P-h diagram for showing the status of pressure in the system.

Mass and energy balance of heat exchanger#1:

$$\text{Mass of air: } m_{ai} = m_{ao} = m_a \quad (2)$$

$$\text{Mass of condensed moisture: } m_w = m_a(\omega_1 - \omega_2) \quad (3)$$

Energy balance for air flow over the cooling coil of heat exchanger#1:

$$m_r(h_2 - h_1) = m_w h_w + m_a(h_5 - h_6) \quad (4)$$

$$h_w = C_{pw}(T_7 - T_0) + h_{lv}$$

Similarly mass and energy balance of compressor and heat exchanger#2

$$m_r(h_2 - h_3) + m_a(h_6 - h_7) = W_c \quad (5)$$

$$m_r(h_3 - h_4) = m_a(h_8 - h_7) = Q_r \quad (6)$$

Further, Exergy destructions of all components can be calculated by following equations:[38]

Exergy destructions for a compressor can be calculated by:

$$I_{comp} = [C_p(T_3 - T_2)] - [T_0 C_p \ln(T_3/T_2)] + W_c \quad (7)$$

Exergy destructions for a condenser can be calculated by:

$$I_{cond} = [C_p(T_4 - T_3)] - [T_0 C_p \ln(T_4/T_3)] + [C_p(T_4 - T_3) \ln(1 - T_0/T_4)] \quad (8)$$

Exergy destructions for an evaporator can be calculated by:

$$I_{evp} = [C_p(T_2 - T_1)] - [T_0 C_p \ln(T_2/T_1)] + [C_p(T_2 - T_1) \ln(1 - T_0/T_2)] \quad (9)$$

Exergy destructions for a compressor can be calculated by:

$$I_{exp} = [C_p(T_1 - T_4)] - [T_0 C_p \ln(T_1/T_4)] \quad (10)$$

Experimental Procedure: The device is installed in open space under shed area at Amity University Noida India campus for Month July 2023 to September 2023. The climate conditions in these months are hot and humid just like the climate conditions of Chennai, Mumbai, Kolkata, Visakhapatnam. Therefore, experiments were carried out in these months to know the performance of the device in humid conditions. The various data such as power input to compressor and fans, air flow velocity, inlet and outlet humidity of air, surface temperature of at various locations of tubes, Ambient temperature, refrigerant pressures, and water collection were recorded regularly every 20 minutes. Based on the data collection, energy and exergy analysis are carried out for flow over the moist air and vapor compressor refrigeration cycle for knowing exergy destruction, efficiency, and cost of water production.

b. Solar Powered Peltier Cooling based Water Generator

Figure 5 (a-b) illustrates the schematic diagram and photograph of experimental setup. The setup comprises two identical aluminum condenser plates attached to two peltier devices all held by an acrylic framework in symmetry. Moist air enters the system through an inlet and moves through a series of ducts when it reaches the condensing chamber. Here, the aluminum plates serve as a condensing surface for dehumidification of moist air with the peltier devices affixed to provide the required cooling effect. The principle of peltier effect lies in the thermoelectric phenomenon generated from the passage of electric current through two different semiconductors; the current causes one side of the device to absorb heat while the opposite side releases heat, causing a cooling effect, the aluminum plates are in contact with the colder side of the device. By the use of this principle, the condenser plates are cooled, enabling dehumidification of air through transfer of heat onto the condenser and formation of water droplets on the plate which eventually accumulate at the base of the device and siphoned into a water collection tank.

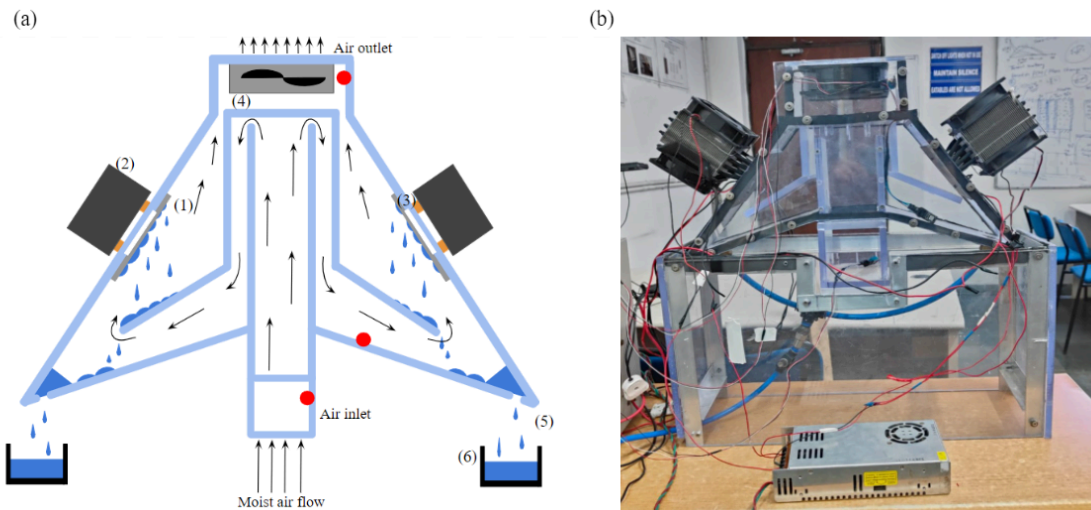


Figure 5: (a) Schematic diagram of Peltier Cooling based Water Generator with position of humidity sensors for measuring the relative humidity and (b) Photograph of experimental setup.

● Position of humidity sensors for measuring the relative humidity at various locations.

(1) Aluminum plate, (2) Heat sink, (3) Peltier device, (4) Exhaust fan, (5) Water outlet, (6) Water collecting tank. Both condenser plates are identical in shape and size.

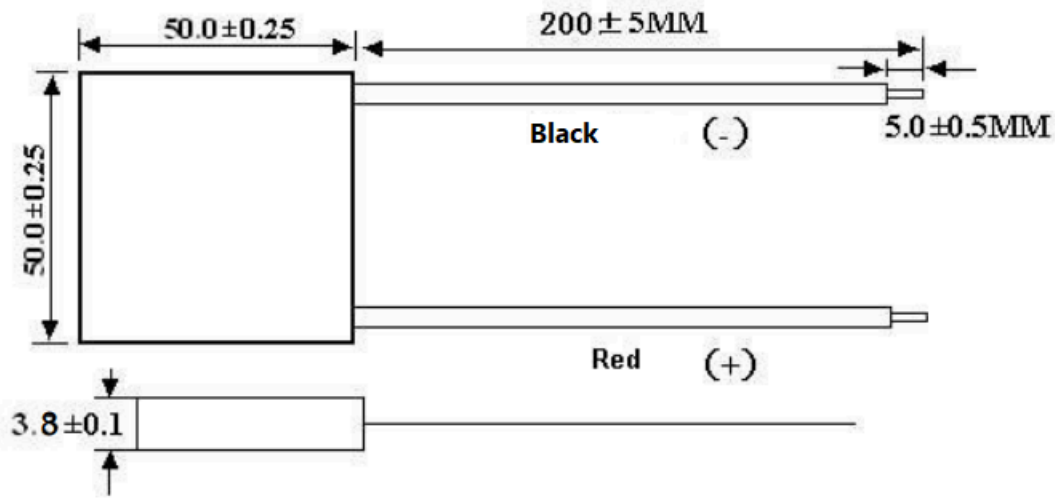


Figure 6: Drawing with dimensions of the peltier device which is used in this apparatus for the cooling effect required for the condensation.

To further increase the effectiveness of the condensers, the aluminum plates are coated in a hydrophobic material of PFOCTS (Perfluorooctyl trichloro silane), this is done by chemically etching the surface with PFOCTS coating under low surface energy conditions. A fan is fastened on top of the apparatus at the outlet to allow the flow of moist air through the system by forcing it through a series of ducts into the dehumidification chamber and the remaining dry air is ejected out of the system through the exhaust fan present at the outlet.

The detailed specifications of all components with their photographs are given in Table 1.

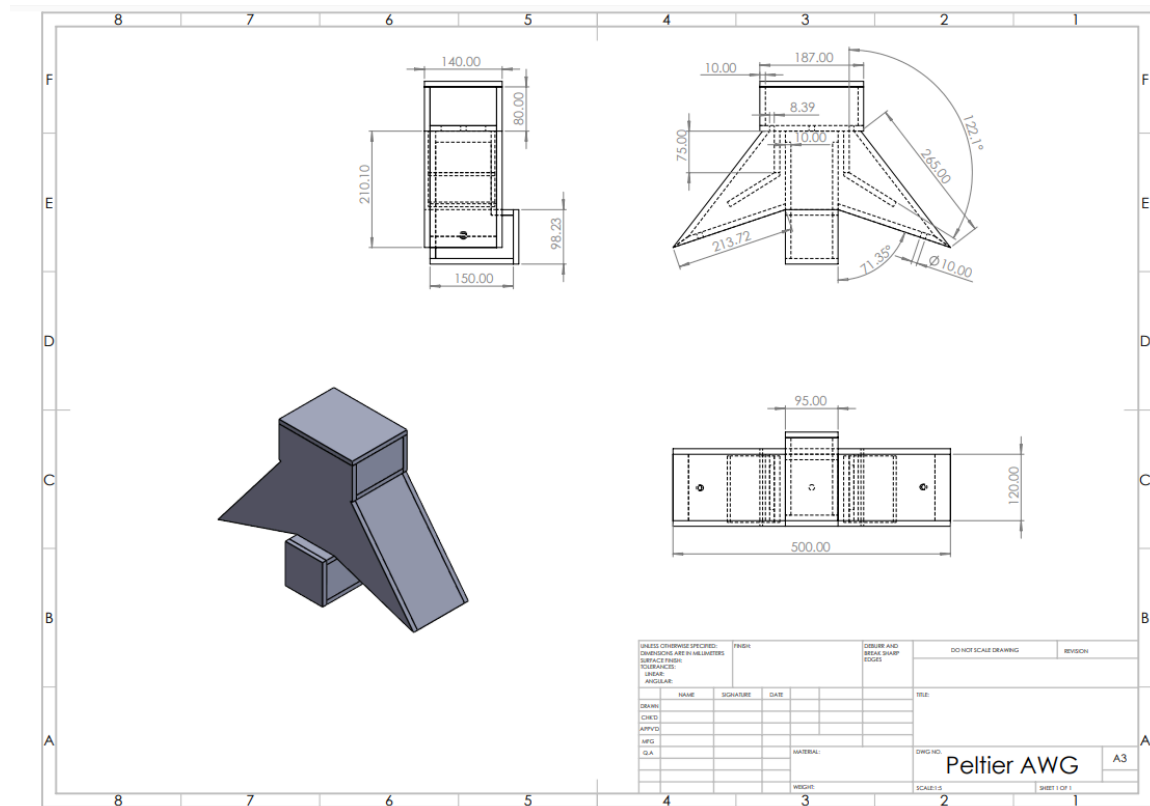
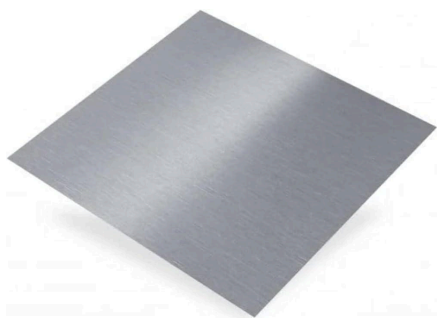
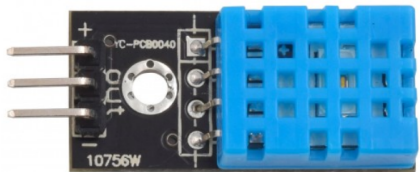
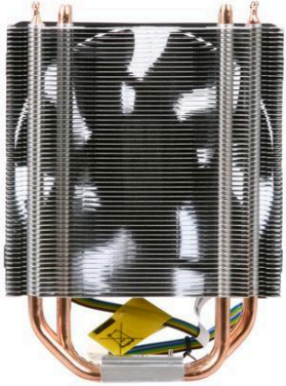


Figure 7: Drawing with dimensions of the apparatus designed and tested.

Table 2: The components of experimental setup with their specification and photographs

Name of components with specification	Photographs
Peltier device: -Model No.: TEC1-12715 -Maximum Operating Temp (T): $<90^{\circ}\text{C}$ -Maximum Cooling Power (Q_{max}): 150 W -Maximum Temp difference (ΔT_{max}): 70°C -Maximum Input Voltage (V_{max}): 16.0 V -Maximum Current (I_{max}): 15.0 Amps -Resistance (R): $0.75 \pm 0.05 \Omega$ -Contact material: BiSn	A photograph of a TEC1-12715 Peltier device. The device is a small, rectangular, grey component with the model number 'TEC1-12715' printed on it. It has two wires extending from the bottom: a red wire and a black wire, which are twisted together.

Condenser Plate: -Material: Aluminum -Area: 100 x 100 mm -Thickness: 2 mm	
Temperature and humidity sensor: -Model No.: DHT11 -Measurement range (Rh/T): 20-90% Rh / 0-50°C -Humidity accuracy (Rh): $\pm 5\%$ Rh -Temperature accuracy (T): $\pm 2^{\circ}\text{C}$ -Type: 4 pin single row -Power supply (V): 5V	
Exhaust fan: -Model No.: SE-12038ASL -Manufacturer: SIBASS -Power (W): 23W -Voltage range (V): 180-220V -Fan Speed: 2600RPM -Current rating: 0.14A -Dimensions: 120mm X 120mm X 38mm	
Water collection measuring beaker: -Capacity: 500 ML -Material: Glass	

Thermal heat sink: -Model: Cooler Master Hyper 212 Plus -Fan speed: 2000 RPM -Control area: 21.2 - 76.8 CFM -Construction: Aluminum -Fan dimensions: 120 x 120 x 25 mm -Heat sink dimension: 116 x 51 x 159 mm -Power connector: 4 pin type	
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The condensed water from the plates is collected in the measuring tank during the experiment. DHT-11 sensors are used to measure temperature and humidity at 3 different points of the apparatus: at the inlet, in the cooling chamber and at the exhaust. The data is recorded on a computer using a 8-bit microcontroller. To enhance the efficacy of the apparatus, extra heat is removed by the use of a thermal heat sink attached to the peltier devices.

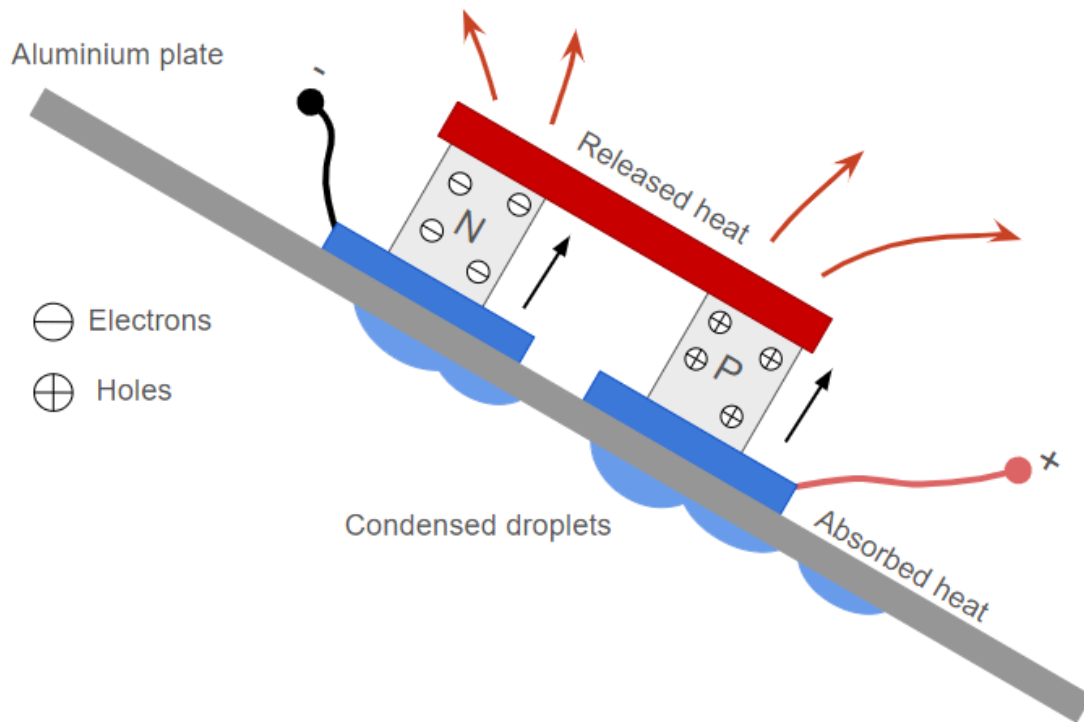


Figure 8: Working of a typical peltier cooler in contact with the aluminum plate for condensation.

Working of Peltier device: The TEC1-12715 thermo-cooler works by arranging N- and P- type semiconductors as a couple, forming a junction via a metallic strip. When a D.C. current is passed through the couple, heat is absorbed at the cold side of the junction and released at the hot side. As a result, there are many such couples placed in series and pressed between substrates such as those of ceramic creating a peltier module.

The heat transportation in a peltier module is done by the charge carrier phenomenon, where electrons (the negative charge carriers) are present in n-type semiconductors and holes (the positive charge carriers) are present in p-type semiconductors. The holes are an empty space that can be occupied by an electron and are free to move about within the lattice, often carrying heat with them. Hence, electrons are responsible for energy release and absorption when provided with current specifically when electrons move to higher energy levels absorb heat and release heat when moved to lower energy levels.

Simulation Procedure for condensation on aluminum plates

The simulation is conducted in ANSYS software using the FLUENT module.

a. *Solution Methodology*- for a numerical analysis, a 2D model was generated in ANSYS and analyzed using the FLUENT software. Mesh set to fine grids near the condenser plates and coarse settings within the body. The condenser plates were refined using the edge-sizing method, average room temperature was same within and inside the apparatus. Momentum and energy equations were selected in a first-order upwind approach. SIMPLE (Semi-Implicit Method for Pressure-Linked Equations) algorithm was used to identify pressure areas.

b. *Mathematical Models*- Since the numerical analysis involves transient, time-dependent phase change of vapor-air to liquid water, this involves usage of various equations:

1. Equations of continuity-

$$\partial \rho / \partial t + \partial (\rho u) / \partial x = 0 \quad (11)$$

2. Equations of momentum-

$$\rho [\partial V / \partial t + V \cdot \nabla V] = -\nabla p + \mu \nabla^2 V + \frac{1}{3} (\mu \nabla (\nabla \cdot V) + \rho g_i \quad (12)$$

3. Energy Equation-

$$\partial (\rho e_i / \partial t) + \nabla \cdot [V \rho e_i + p] = \nabla \cdot [k \nabla T + (\tau \cdot V)] + S_g \quad (13)$$

For modeling turbulence occurring between phase change, the viscous model of standard k- ϵ was chosen with enhanced wall treatment.

c. *Boundary conditions*- to reduce the computational time, half the model was analyzed with symmetry conditions applied. Inlet and outlet is applied to the model to allow the passage of vapor-air through the system where the heat is transferred between the aluminum plates given a lower temperature than the surroundings. To simulate the hydrophobicity of plates, a low wall adhesion coefficient is given and contact angle is set high.

d. *Modeling of the condensation*- initial conditions are set for vapor-air to entertain the system at 80% humidity at 30°C with backflow volume fraction set to 1.

Condensation on a wettability of the surface

The condensation process is essential for transferring large amounts of energy using the latent heat during a phase transition. In the condensation of vapor, a species mass transfer phenomenon occurs where vapor moves to the liquid-vapor interface and converts to liquid, this transport is necessitated by a pressure difference (from higher to lower) already present at the interface.[45] As such, condensation on any surface is termed as “heterogeneous condensation” occurring in two modes: formation of a continuous thin, liquid film known as filmwise condensation (FWC) or by formation of droplets over the surface that simply slides off the surface, this is known as dropwise condensation (DWC)[46].

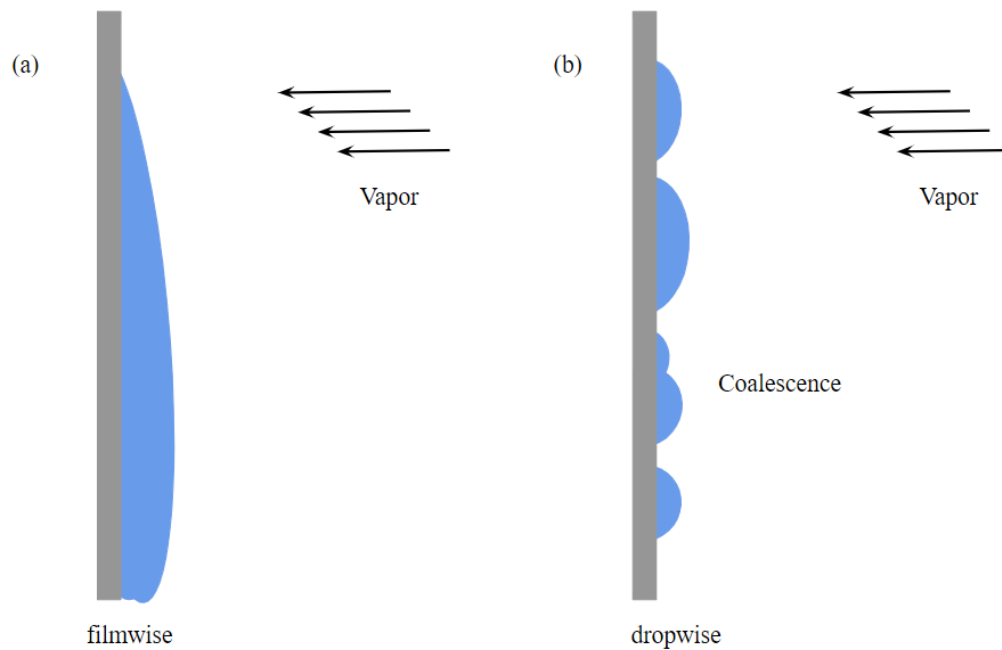


Figure 9: Filmwise and dropwise condensation.

The type of formation formed depends upon the surface properties of the condensing medium particularly that of cohesive and adhesive forces between the solid surface and liquid which further determines the “wettability” of the surface, wettability is the preference of liquid to be in contact with a solid surrounded by another liquid.

Filmwise condensation (FWC) involves formation of a thin film that covers the entire surface which causes a high thermal resistance between the vapor-liquid interface, thus, this slows down the mass transfer rate over the course of time.[47] The stability of the liquid film during condensation is influenced by the interplay of shear stress, gravity, and

surface tension. At low vapor velocities and subcooling (the temperature difference between the surface and the saturation temperature), a laminar film with smooth flow is formed, such surfaces need to have high wettability for such films to form.[48] Conversely, rough or non-wetting surfaces hinder the formation and movement of the liquid film. [49]

Dropwise Condensation (DWC) occurs when the vapor to liquid phase change forms into discrete drops over the surface[47], the condensate does not wet the surfaces except at locations where a wetted contaminant nuclei exists, this allows for higher heat coefficient during the process as compared to Filmwise Condensation (FWC). Condensation occurs at preferred nucleation sites at atomic scales and keeps growing up to the point where size of the droplet is equal to the distance between other neighboring nucleation sites, at which, the droplets coalesce to form bigger drops. This process of condensation and coalescence continues till the body force exerted by the drop exceeds the surface tension of the plate-drop interface.[50] Such surfaces need to have low wettability for drop formation to occur, however, for such conditions to cause droplet formation the wettability characteristic must also determine the contact angle between the liquid and surface interaction. For any given liquid between different substrates, there are three main types of angles: hydrophilic ($0 \leq \theta \leq 90^\circ$), hydrophobic ($90^\circ \leq \theta \leq 140^\circ$) and superhydrophobic ($140^\circ \leq \theta \leq 180^\circ$) [50,51]. There are many methods to achieve such hydrophobic surfaces by deposition method through chemical etching forming an appropriate substrate. [39,40]

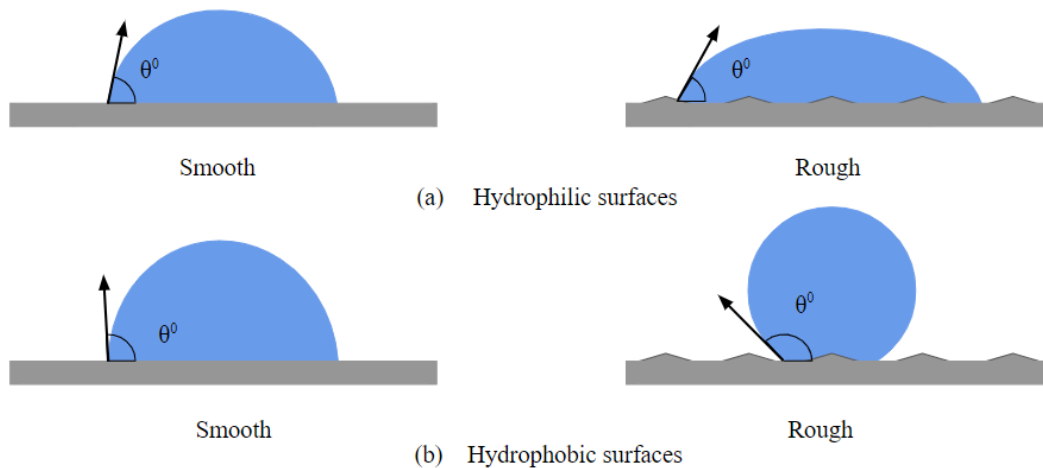


Figure 10: Measurement of different contact angles on different substrates.

Results and Discussion

The water collection rate, power consumption and effective cooling air flow compressor and effect of environmental conditions on water generation rate were present in this section. The compressor was present in this section. Figure 5 shows the water collection in the months of 16th July 17th and 18th August, and 4th September. All reading was taken every 20 minutes in a day. These Figures show that the water collection varies morning to evening because of variation of relative humidity.

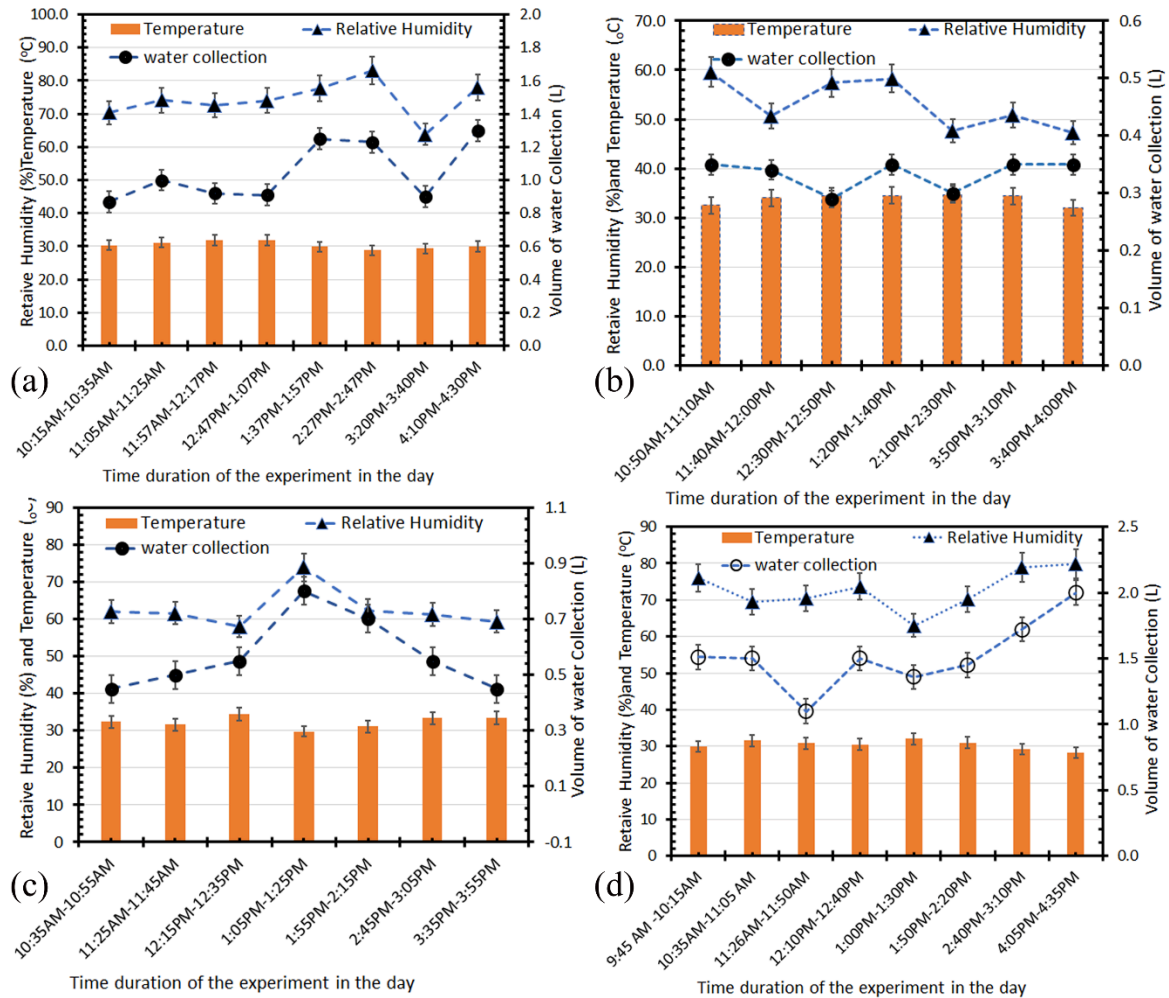


Figure 11: Variation of water collection and environmental conditions with respect to duration in the day in Noida India. (a)18th July 2023, (b)16th August,2023 (c) 21st August 2023 and (d) 16th September 2023.

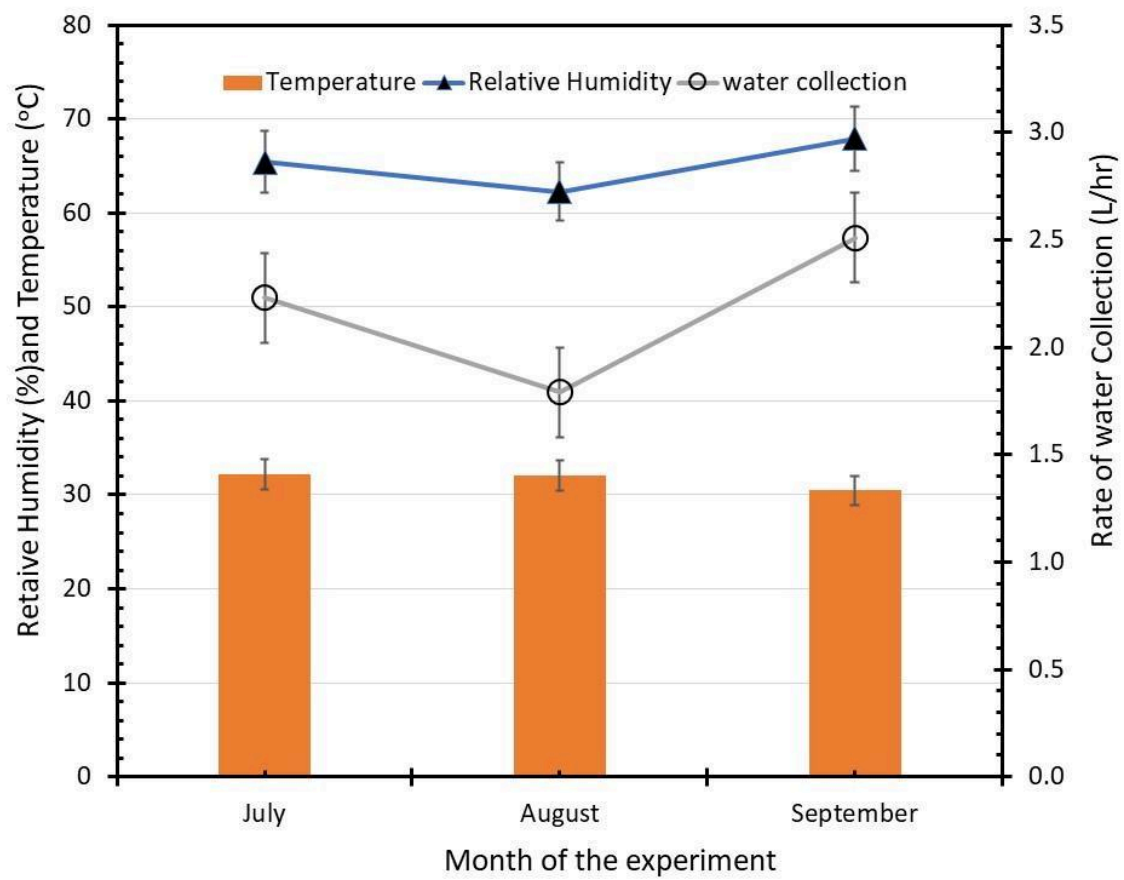
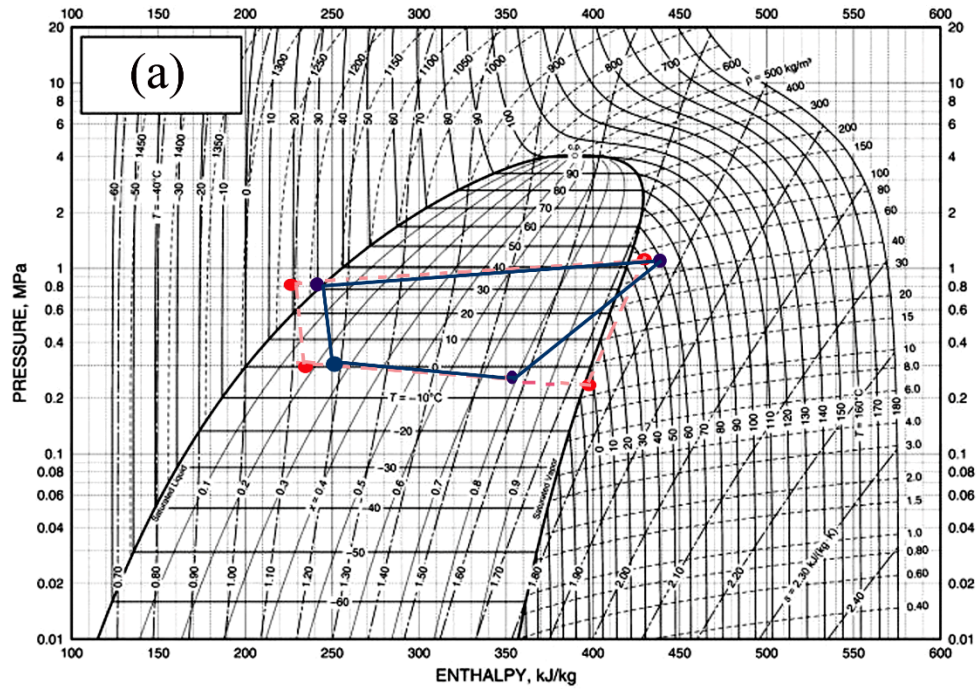
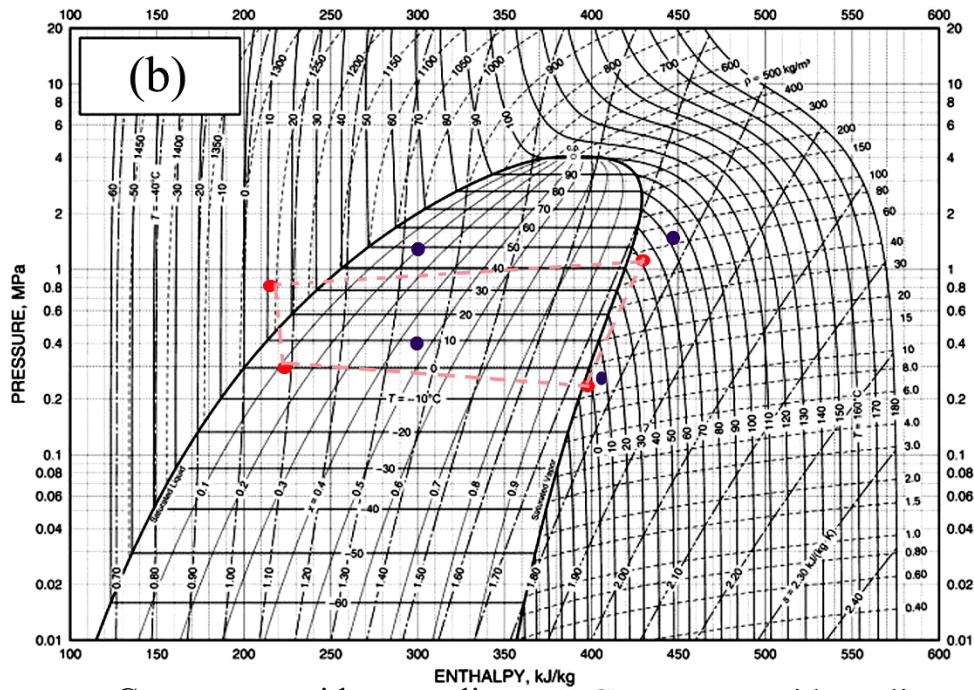


Figure 12: Capacity of water production and Power consumption at various conditions in the month of July-September 2023 in Noida India.



- Cooling cycle for plane tubes
- Cooling cycle for hydrophobic surface



- Compressor without cooling
- Compressor with cooling

Figure 13: Vapor compression cycle in P-h diagram during water vapor condensation in moist air. (a) The outer surface of heat exchanger#1 is normal (hydrophilic: without coating) and modified (hydrocooling coating) (b) compressor with flow of cooling air and without flow of cooling air. The experiment (a) was conducted on 5th September 2023 and experiment (b) was carried out on 7th September 2023.

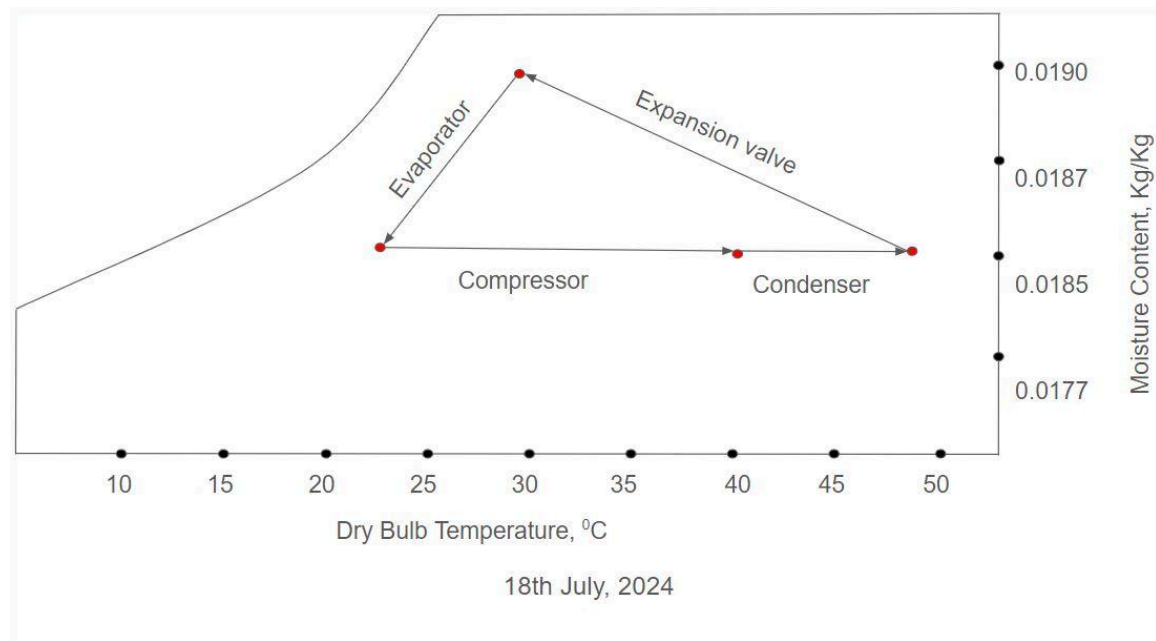


Figure 14: Vapor compression cycle in Psychrometric chart during water vapor condensation of moist air, readings taken on 18th July, 2023.

Table 3: Environmental Conditions, water collection rate, power consumption and cost of water table.

		Environmental Conditions					
S.no	Day	Average Velocity (m/s)	Average Temperature (°C)	Average Humidity (Rh%)	water collected (l/hr)	Power consumed (kW)	Cost of water (Rs/L)
1	18 th July	4.4	29.02	74.2	3.14	20.65	6.16
2	19 th July	4.6	28.61	62.29	2.06	20.72	9.42
3	31 st July	4.7	30.56	59.77	1.49	17.55	12.65
4	2 nd August	4.8	31.16	61.75	2.17	20.39	10.47
5	3 rd August	4.8	33.43	60.07	1.75	18.27	11.85
6	14 th August	4.7	31.58	59.81	1.49	20.65	14.44
7	16 th August	4.7	30.45	62.63	1.48	18.24	11.40
8	17 th August	4.4	30.58	53.1	1.15	18.21	19.53
9	23 rd August	4.5	26.82	81.34	3.64	13.10	5.41
10	28 th August	4.7	32.07	53.34	1.24	18.36	15.92
11	13 th September	4.5	32.35	63.1	1.80	12.80	10.63
12	15 th September	4.5	29.43	67.9	7.35	20.82	7.48
13	16 th September	4.6	27.56	72.7	9.34	20.81	6.27

Table 4: Inlet and Outlet conditions of Heat Exchanger#1 and Heat Exchanger#2

Heat Exchanger # 1		Air Inlet Conditions			Air Outlet Conditions		
<u>S.no.</u>	<u>Day</u>	Temp (°C)	Sp. Humidity (kg/kg)	Velocity of air (m/s)	Temp (°C)	Sp. Humidity (kg/kg)	Velocity of air (m/s)
1	18 th July	29.02	0.0189	4.4	23.16	0.0185	4.4
2	19 th July	28.61	0.0155	4.6	23.67	0.0152	4.6
3	31 st July	30.56	0.0166	4.7	22.54	0.0164	4.7
4	2 nd August	31.16	0.0178	4.8	23.56	0.0176	4.8
5	3 rd August	33.43	0.0198	4.8	21.23	0.0196	4.8
6	14 th August	31.58	0.0177	4.7	22.47	0.0175	4.7
7	16 th August	30.45	0.0173	4.7	23.08	0.0171	4.7
8	17 th August	30.58	0.0147	4.4	25.46	0.0146	4.4
9	23 rd August	26.82	0.0183	4.5	23.45	0.0179	4.5
10	28 th August	32.07	0.0162	4.7	24.56	0.0160	4.7
11	13 th September	32.35	0.0196	4.5	24.16	0.0193	4.5
12	15 th September	29.43	0.0177	4.5	23.78	0.0168	4.5
13	16 th September	27.56	0.0171	4.6	21.54	0.0159	4.6

Heat Exchanger # 2		Air Inlet Conditions			Air Outlet Conditions		
<u>S.no.</u>	<u>Day</u>	Temp (°C)	Sp. Humidity (kg/kg)	Velocity of air (m/s)	Temp (°C)	Sp. Humidity (kg/kg)	Velocity of air (m/s)
1	18 th July	40.76	0.0185	4.4	49.37	0.0185	4.4
2	19 th July	40.51	0.0152	4.6	45.73	0.0152	4.6
3	31 st July	40.73	0.0164	4.7	48.78	0.0164	4.7
4	2 nd August	40.38	0.0176	4.8	47.52	0.0176	4.8
5	3 rd August	40.15	0.0196	4.8	53.47	0.0196	4.8
6	14 th August	40.28	0.0175	4.7	49.67	0.0175	4.7
7	16 th August	41.27	0.0171	4.7	47.67	0.0171	4.7
8	17 th August	41.19	0.0146	4.4	45.43	0.0146	4.4
9	23 rd August	40.58	0.0179	4.5	44.45	0.0179	4.5
10	28 th August	41.24	0.0160	4.7	48.76	0.0160	4.7
11	13 th September	44.38	0.0193	4.5	52.59	0.0193	4.5
12	15 th September	39.97	0.0168	4.5	46.23	0.0168	4.5
13	16 th September	41.04	0.0159	4.6	47.34	0.0159	4.6

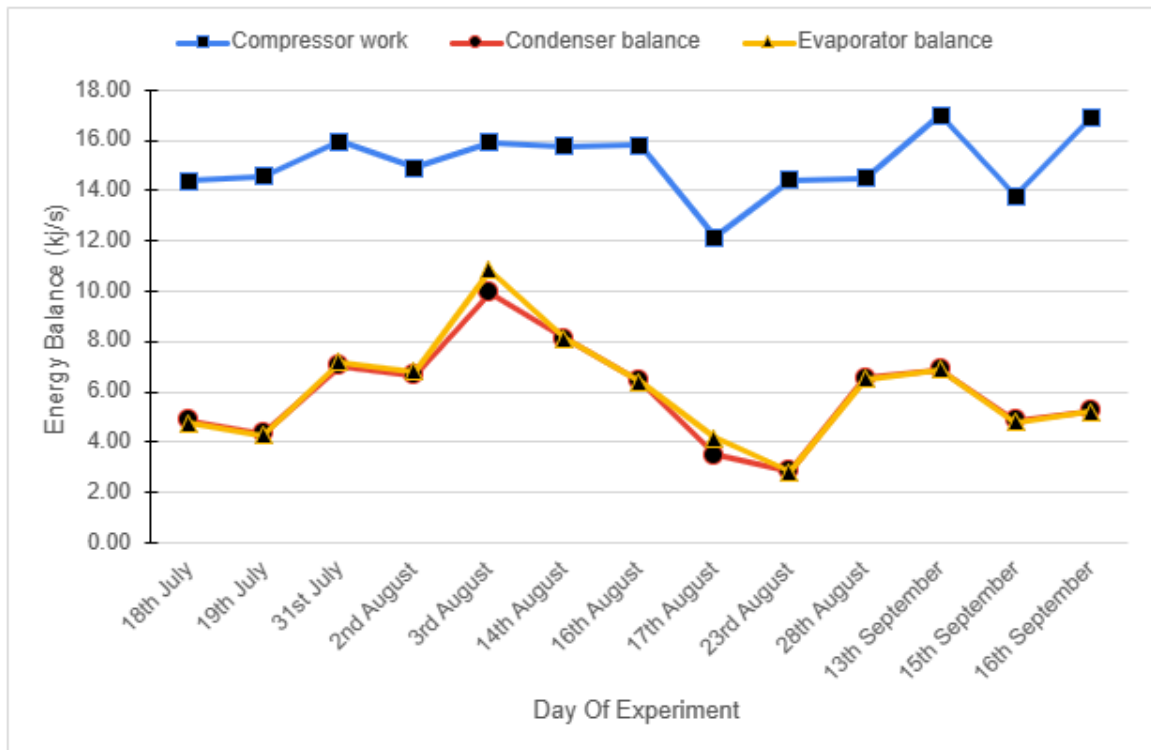


Figure 14: Energy Balance chart on different days for different components present in the VCRS system.

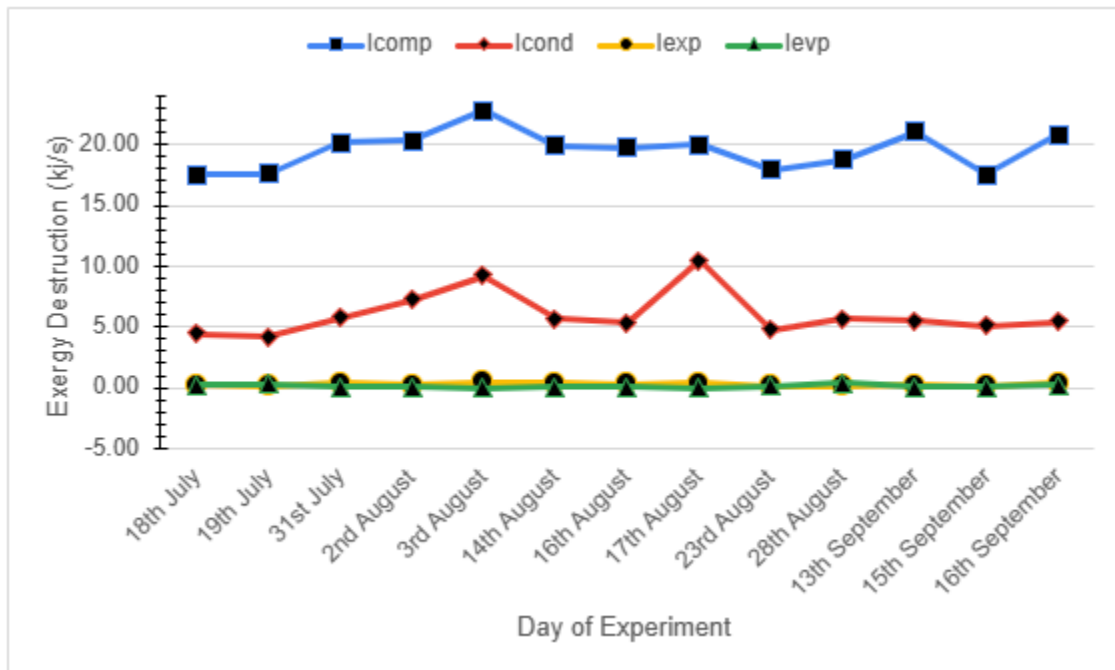


Figure 15: Exergy Balance chart on different days for different components present in the VCRS system

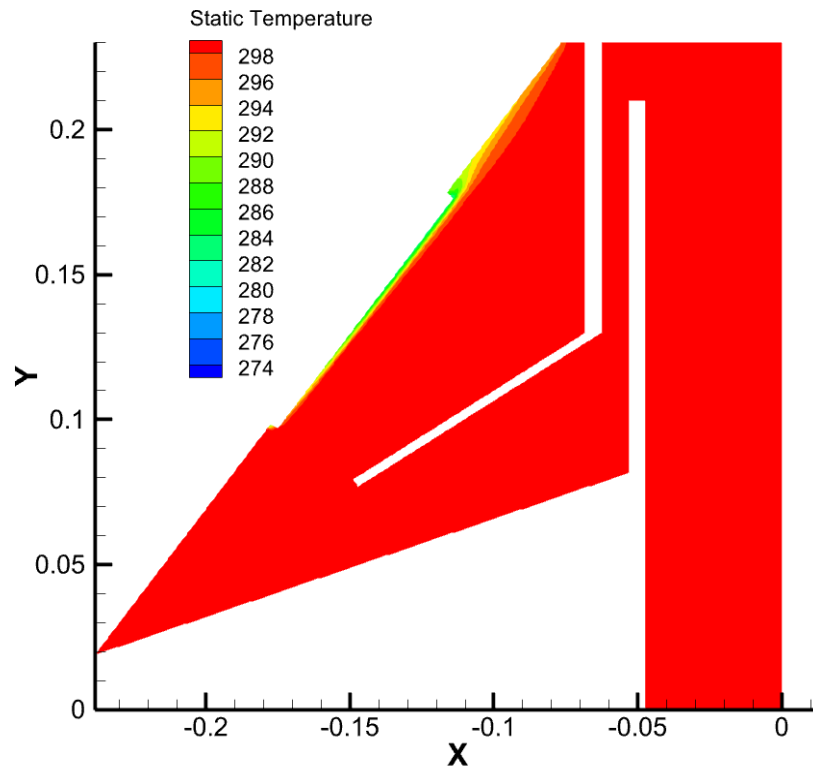


Figure 16: The temperature contour of the air flowing through the system.

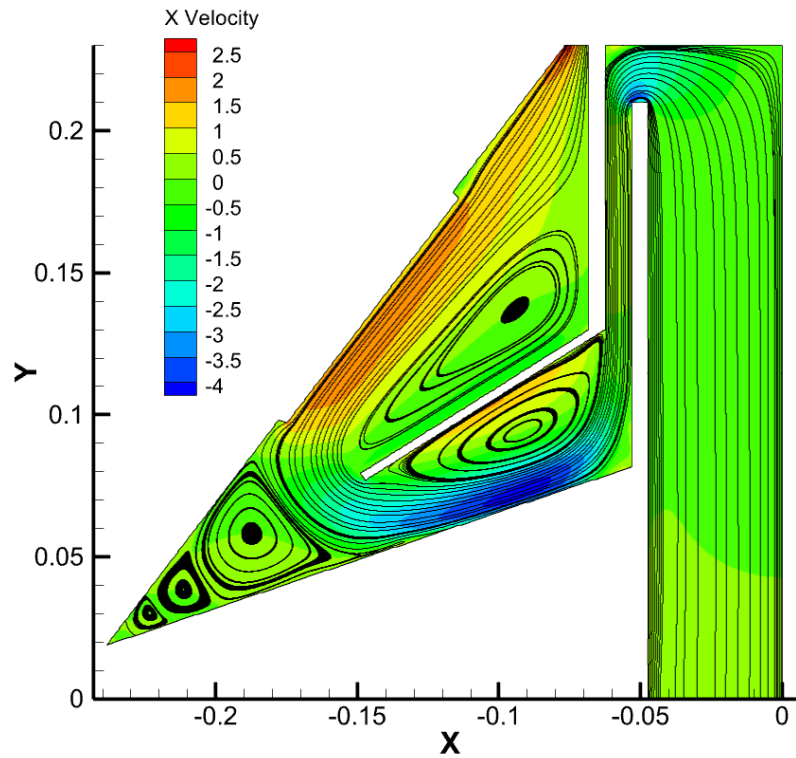


Figure 17: The velocity contour and sketch lines of the air flowing through the system.

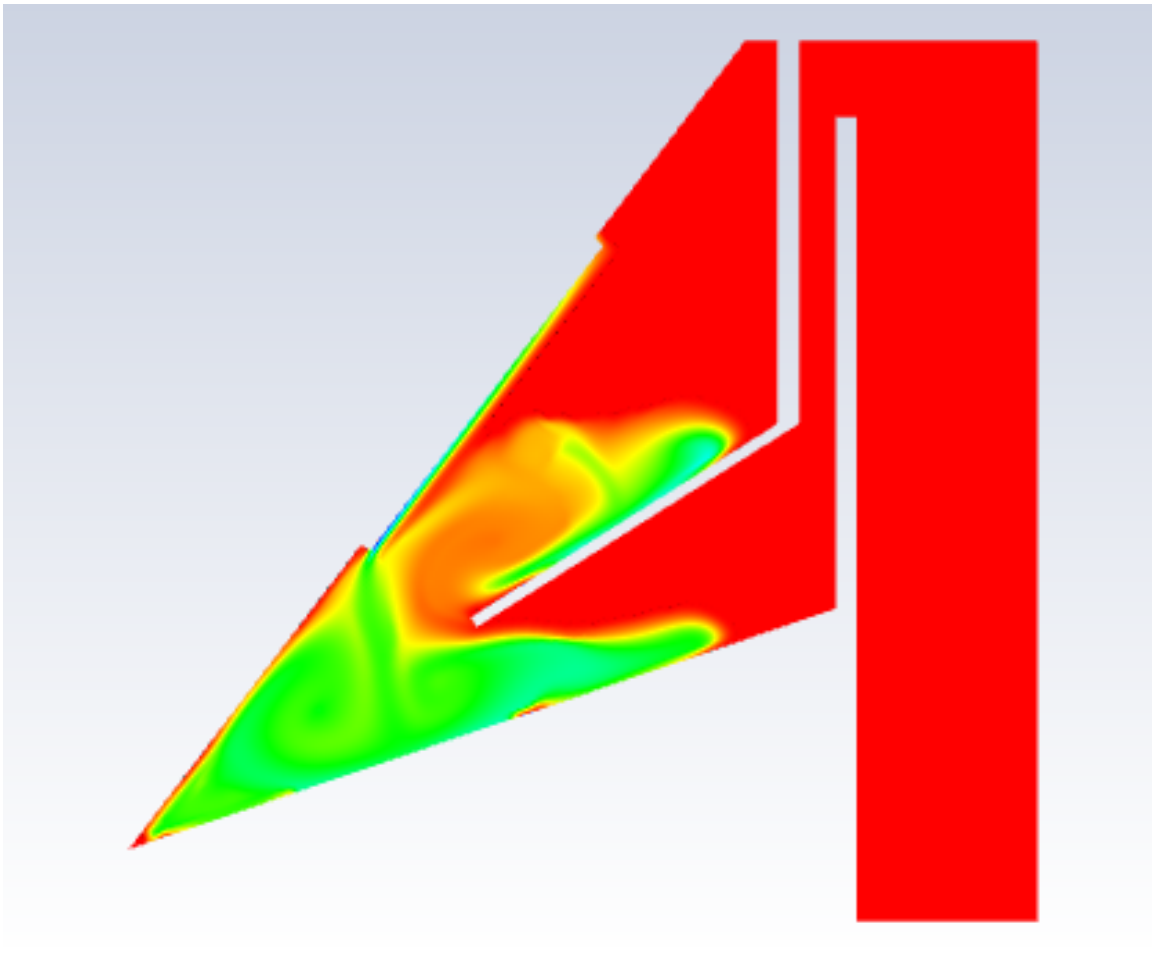


Figure 18: condensed water formation on the condenser plate of the apparatus.

Summary and conclusion

Atmospheric water generator is widely used to produce water from the air in arid and humid regions. In this work, a vapor compression cooling based atmospheric water generator is designed and fabricated and thermodynamic analysis was carried out to understand its efficiency and economic feasibility. The experiments are carried out at actual environmental conditions for three months (July-September 2023) in Noida India. Temperature at inlet and outlet of compressor as well as inlet outlet of condensing surface were measured. In addition, power consumption rate and rate of condensate were recorded at various environmental conditions. Using recorded data, exergy, energy, and mass balance estimated of each component as well as the complete system. This study observed that the proposed device condensed 10 liters of water vapor per hour at a temperature of 32°C and Relative humidity of 70%. However, the cost of water is 4 INR per liter at minimum destruction of exergy. Hence, this research contributes to a deeper understanding of vapor compression based atmospheric water generator system thermodynamics, offering insights crucial for optimization and further development of this technology as a sustainable water generation solution.

further, a CFD study was conducted on the Peltier apparatus, testing the flow of air through the system and the condensation of droplets over the aluminum plates. The following conclusion is drawn from this study.

- The atmosphere has sufficient water for resolving potable water scarcity on earth if cooling of condensing surface is effective and sufficient for moist air condensation.
- In this work, the AWGs are designed and fabricated for producing water from the atmospheric air. However, cooling of condensing surface is based on vapor compression and condensing surface is hydrophobic copper tubes ,while for other apparatus, cooling is done via peltier coolers using hydrophobic aluminum plates as the condensing surface.
- The experiment studies were carried out, and condensate were collected at real environmental conditions in the months of July to September 2023 in NOIDA, India.

- This research contributes to a deeper understanding of vapor compression based atmospheric water generator system thermodynamics, offering insights crucial for optimization and further development of this technology as a sustainable water generation solution.

Acknowledgment

Apoorva Singh acknowledges the Science and Engineering Research Board (SERB), Government of India (Project number CRG/2021/005669) for providing funds for carrying out this work. Amity School of Engineering and Technology Noida, India is acknowledged for providing a high-performance computing platform.

Future work

Future research will simulate and validate the condensation of vaporized air using surface coolers, testing the aluminum plates hydrophobicity and condensation rate. Based on the simulation that was done, the apparatus may be further improved, and any defects in the current workpiece may also be examined.

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Major Project Report

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Approva Singh

WORD COUNT

6344 Words

CHARACTER COUNT

36687 Characters

PAGE COUNT

34 Pages

FILE SIZE

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